



Laser joining of metals to plastics with ultrasonic vibration



Y.J. Chen^{a,*}, T.M. Yue^a, Z.N. Guo^b

^a The Advanced Manufacturing Technology Research Centre, Department of Industrial and Systems Engineering, The Hong Kong Polytechnic University, Hong Kong

^b School of Electromechanical Engineering, Guangdong University of Technology, Guangzhou, China

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ABSTRACT

Ultrasonic-aided laser joining process (UAL) for bonding metals to plastics has been further developed. With the application of an appropriate ultrasonic transducer tool, the quantity of laser-induced bubbles in the bonding zone can be significantly reduced. The removal mechanism of these bubbles from the molten plastic zone was analyzed based on pressure induction on the sample under ultrasonic vibration. Some experiments on joining titanium to polyethylene terephthalate have been conducted to ascertain the superiority of UAL over the conventional laser joining process. The joint strength, measured in terms of failure stress, was significantly increased when ultrasonic vibration was employed during laser joining. The joints formed without vibration always failed by separation along the metal-plastic interface, whereas, for the UAL joints, fracture occurred in the parent plastic material.

1. Introduction

In recent years, considerable interest has been shown in joining dissimilar materials such as light metals to plastics, especially in the automobile (Grujicic et al., 2008) and biomedical industries (Sutana et al., 2009). The demands of integrating different materials are obvious, due to energy saving, environmental concern, as well as the requirements for high functionality of light weight products. Many automobile and biomedical products are made of multi-components where various types of materials are used so the advantages of each material are realized. The joining of plastics to metals is not new; conventionally, joining methods consist of adhesive bonding (Elena and Lazar, 2003) and mechanical fastening by screws or rivets (Amancio-Filho and dos Santos, 2009). These traditional techniques have their own drawbacks; joining polymers to metals using conventional methods is normally difficult because of the incompatible structural, physical and chemical properties between these two types of materials. To circumvent some of these problems, Katayama and Kawahito (2008) developed a direct laser bonding technique for joining metals to plastics, namely laser-assisted metals and plastics joining (LAMP). Other welding techniques also show potential for joining metals to polymer-based materials, such as friction-based lap welding and spot joining (Liu et al., 2014; Amancio-Filho et al., 2011; Abibe et al., 2016), induction (Mitschang et al., 2013) and ultrasonic (Wagner et al., 2013; Yeh and Hsu, 2016) welding. Similar to LAMP, friction-based techniques also produce bubbles in the bond area due to decomposition of plastic material. Induction spot welding is a relatively rapid process and shows

high reproducibility and reliability, although it produces metals/carbon fibre reinforced plastics joints with strength slightly less than that formed by using adhesive bonding. In a comparison of the various joining techniques for forming metal-plastic hybrid joints, ultrasonic welding has shown to be superior, in terms of shear strength (Amancio-Filho et al., 2011; Wagner et al., 2013). To increase the strength of ultrasonic formed hybrid joints, an attempt was made to employ laser surface-structured metals in ultrasonic joining (Yeh and Hsu, 2016).

LAMP has been employed to rapidly produce strong joints between commercially available metal alloys and engineering plastics (such as titanium/PET (Chan and Smith, 2016), zinc-coated steel/ABS (Jung et al., 2016), and aluminum/PET (Yusof et al., 2012)). LAMP involves laser beam transmission through the plastic that leads to a sufficiently high temperature rise at the interface between the metal and the plastic material. The plastic in the joint zone is melted and decomposition occurs; this causes the formation of some small bubbles in the molten plastic, which remain in the solidified plastic. After the molten plastic has cooled down, adhesive bond with high shear strength, was formed between the metal and the plastic (Katayama and Kawahito, 2008). It is generally accepted that the laser-induced bubbles induce high pressure in the molten plastic and force it into intimate contact with the metal surface, whereupon the molten plastic enters the concavities to accomplish anchoring effects (Wahba et al., 2011). Yusof et al. (2012) found that the interactive force of the chemical bond and van der Waals forces are also thought to play an important role in securing a strong joint. Turning to the effect of laser processing parameters on bubble formation, using a low welding speed with a high power input (32 mm/

* Corresponding author.

E-mail addresses: 13902125r@connect.polyu.hk (Y.J. Chen), tm.yue@polyu.edu.hk (T.M. Yue), znguo@gdut.edu.cn (Z.N. Guo).

min; 80 W) resulted in a large amount of sizeable bubbles and low joint strength (Tillmann et al., 2010). The amount of bubbles formed in the joint has a significant impact on joint strength, which is in agreement with the findings of Cheon and Na (2014). The research results thus far have shown that the size, amount and the distribution of the bubbles are the key factors that directly affect the joint performance (Tamrin et al., 2013). It is clear that the laser-induced bubbles, although assisting in creating an intimate contact surface between the metal and plastic parts by inducing high pressure in the molten plastic, are themselves defects by nature, and therefore have a counter effect on the joint strength (Tillmann et al., 2010). Until now, there is still no effective method that can be used to disperse the large bubbles into smaller ones or to greatly reduce the laser-induced bubbles.

Recently, Chen et al. (2016) have developed an ultrasonic-aided laser joining method (UAL) that has been shown to be superior to LAMP. The improved process has been demonstrated by a study of joining polyethylene terephthalate (PET) to pure titanium (Ti ASTM B265 Grade 1), under a low laser power condition (30 W). The formation of bubbles was largely suppressed because a low laser power was used to avoid decomposition of the polymer, and only a few small bubbles (less than 85 μm) were found, under such a condition, the laser-induced bubble problem was not a serious concern. This means that lowering laser power can avoid decomposition of the polymer and formation of a large number of bubbles. Moreover, because of better surface contact, high joint strength can be obtained. The action of ultrasonic vibration also causes interfacial friction and a volumetric viscoelastic heating effect, which enhance chemical reaction. As a result, a stronger joint interface with a higher shear failure load is obtained. Building on the success of this previous study, the present study aims to further develop UAL to address the laser-induced bubble problem, with the aim of reducing bubbles at the plastic-metal joint interface when relatively high laser powers (45 W–75 W) are used.

2. Experimental method

2.1. The ultrasonic-aided laser joining method

UAL was performed using a specially designed ultrasonic-aided laser joining system (Fig. 1a). The system essentially consisted of a 300 W pulsed Nd:YAG laser which has a multi-mode beam shape, an ultrasonic vibration system capable of vibrating at a frequency of 28 kHz and a clamping device. The ultrasonic vibration system consists of a transducer tool on which the workpiece (PET and Ti) rests (Fig. 1b). The tool was made of Cr12 steel. Two different surface profiles were used for the tool. One has a flat surface with circular blind holes (Fig. 2), and the other has a flat surface with an open channel (Fig. 3). The purpose of having cavities in the transducer tool is to create a pressure field in the joint specimen when ultrasonic vibration is applied. The differential pressure effect would cause bubbles to move in the molten plastic pool, and with a proper tool design, the bubbles can be made to escape from the joint zone. The blind-hole-design aims to create a relatively low pressure region in the molten plastic zone above the location of the holes, and is intended to test the idea that under ultrasonic vibration, bubbles will move to these low pressure regions. A laser Doppler vibrometer (Polytec, PSV-400) was used to measure the vibration amplitude of a high pressure point (Position A) and a low pressure point (Position B) of the specimen placed above the transducer tool (Fig. 4) by positioning the meter directly above these two locations. The open-channel-design aims at providing a low pressure region for the bubbles to escape from the molten pool.

The vibration system also includes a clamping device which has a top quartz plate through which the laser beam is transmitted onto the workpiece, with two electrically driven cylinders employed to apply a clamping force to the workpiece (Fig. 1b). The workpiece (the materials to be joined) has the PET sheet placed on the top of the titanium sheet. The laser beam passes through the quartz plate and the PET sheet, and

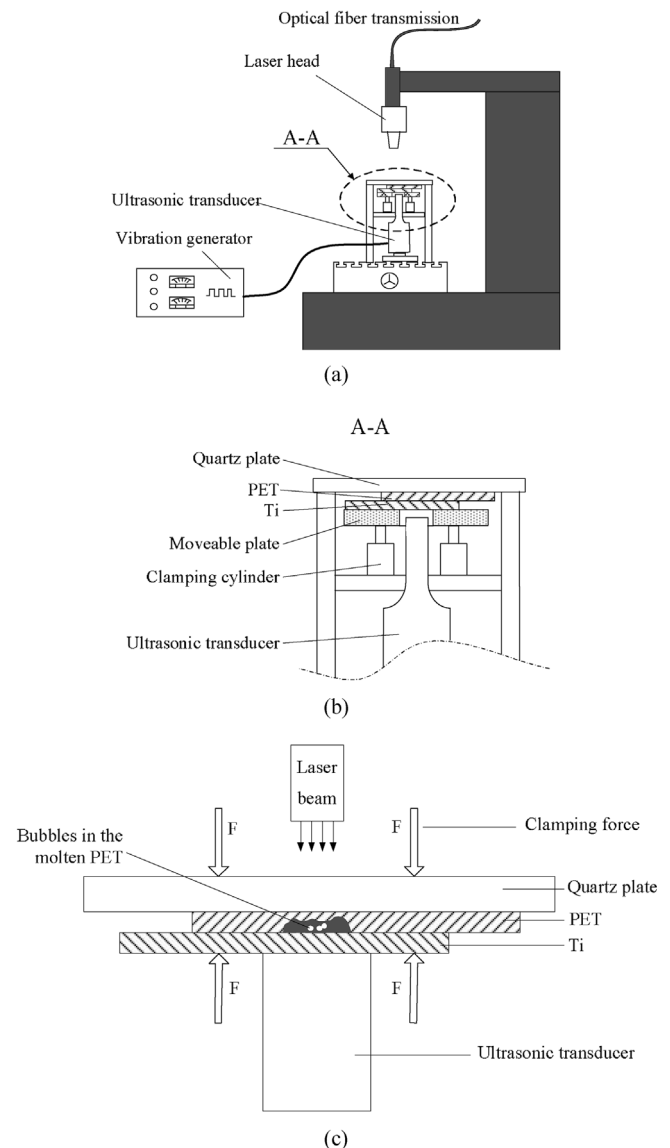


Fig. 1. (a) Ultrasonic-aided laser joining system; (b) more details of A-A; (c) a schematic illustration showing the joining process.

when it reaches the surface of the titanium sheet, due to the absorption of the laser by the Ti metal, it is heated up, causing melting and some degree of vaporization (decomposition) of the contact surface of the PET sheet (Fig. 1c). The joint was formed by using one pass of laser beam only. During laser joining, the ultrasonic device operates at a duty cycle of 20% (duty cycle = ultrasonic operation time/welding time) (Fig. 5). The vibration amplitude was 4 μm because if too low, the differential pressure effect would be relatively weak. On the other hand, the previous study showed that if the amplitude was higher than 4 μm , plastic degradation would occur (Chen et al., 2016). Total welding time was 5 s, and ultrasonic vibration was applied during the welding process, when times were 2 s and 4.5 s, each for a duration of 0.5 s. This produced a lap-joint length of about 20 mm with a joint seam width in the range of 3–5 mm. A wider seam can be produced by moving the transducer along the x-axis incrementally. LAMP and UAL joining parameters used in this study are listed in Table 1, and designations of the specimens are presented in Table 2.

2.2. Materials to be joined

Both LAMP and UAL joining experiments were conducted for a

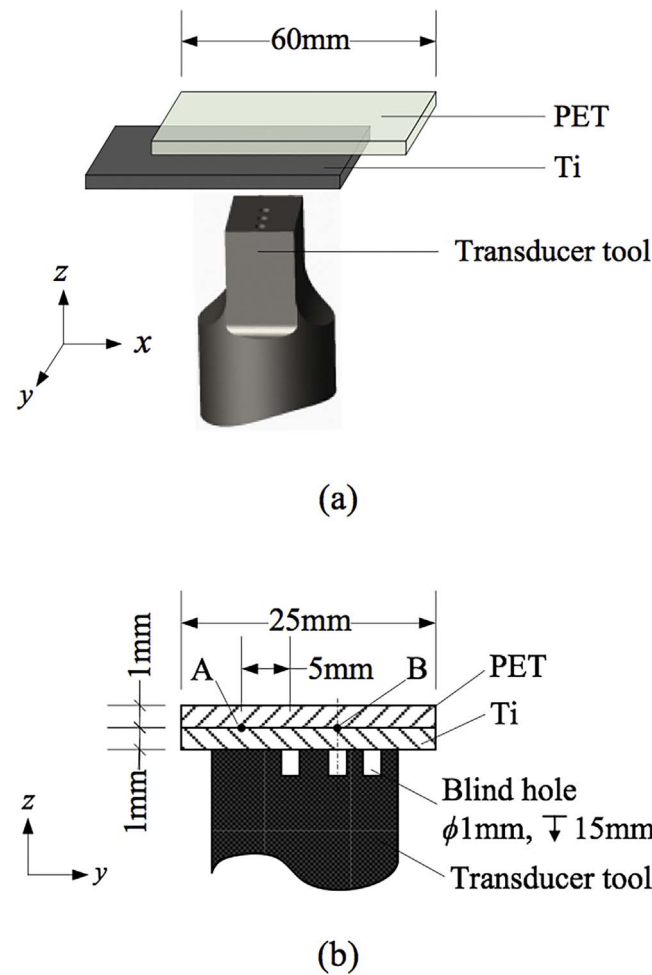


Fig. 2. (a) Ultrasonic transducer tool with circular blind holes, (b) a cross-section along the y-axis.

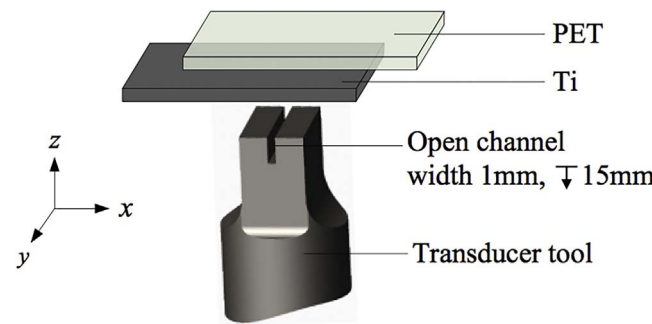


Fig. 3. Ultrasonic transducer tool with an open channel.

titanium (Ti)/polyethylene terephthalate (PET) pair. Titanium and its alloys are widely applied in aerospace equipment and medical applications due to their excellent properties, such as lightness, high strength and excellent corrosion resistance. PET is the plastic material to be joined to Ti. It is a stiff, strong and dimensionally stable material with a low water absorption rate. It also has good gas permeable resistance properties and good chemical resistance, and is an important material for medical devices. Its chemical formula is $(C_{10}H_8O_4)_n$, which contains the ester carbonyl group. Since the carbon atoms in ester carbonyl are active, chemical reactions should readily occur between PET and the metal part to be joined. Both the Ti sheet (ASTM B265 Grade 1, 1 mm thick, purity 99.6%) and the PET sheet (1 mm thick, amorphous) were purchased from Goodfellow Cambridge Ltd. For the purpose of comparison, shear tests were carried out using simple rectangular

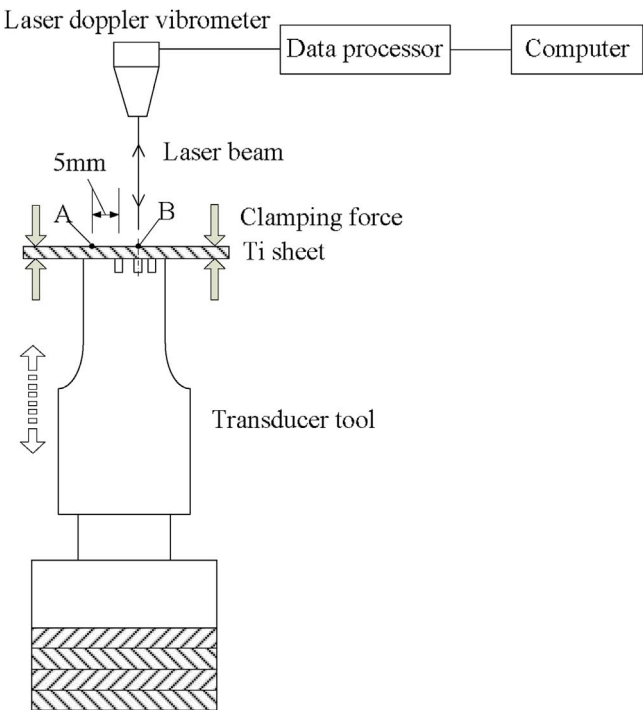


Fig. 4. The laser Doppler vibrometer set up.

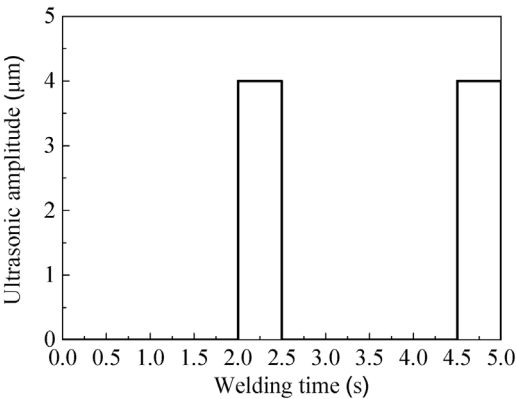


Fig. 5. Ultrasonic pulse operation cycle (duty cycle of 20%).

Table 1
LAMP and UAL joining parameters.

Parameters	Value
Laser focal length	25 mm
Laser power	30 W/45 W/55 W/65 W/75 W
Welding speed	4 mm/s
Laser pulse frequency	20 Hz
Pulse duration	8 ms
Laser spot diameter	φ 2.5 mm
Transducer vibration frequency	28 kHz
Transducer vibration amplitude (only applied to UAL)	4 μm

Table 2
Designations of specimens.

LAMP (without ultrasonic vibration)	LAMP_xx W
UAL (with ultrasonic vibration)	UAL_xx W

xx stands for laser power.

specimens. The Ti and PET strips to be joined have dimensions of 60 mm × 25 mm × 1 mm. The joining face of the titanium specimens was ground finished with emery paper (#320 and #1200) to remove oxide films from the surface, and all the Ti and PET specimens were cleaned in an ultrasonic bath in ethanol for 10 min prior to laser joining.

2.3. Analysis of the joint

The chemical state of the Ti/PET joint interfaces was studied using x-ray photoelectron spectroscopy (XPS) with the XPS analyzer (ULVAC-PHI 1800). To conduct the analysis, some LAMP and UAL joined specimens were separated at the interface by mechanical means, and analysis was performed on the Ti part. The software XPSPEAK was employed to make a semi-quantitative study on the quantity of bonds formed across the joint interface, based on the Ti2p line. To reveal the nature of the chemical bonds across the interface, ion-milling in small steps was employed to thin down and to remove any solidified PET adhering to the Ti surface until reaching the Ti substrate. This was accomplished using XPS analysis after each layer of plastic was removed, until Ti was detected. This level was designated $d = 0$. At this level ($d = 0$), more ion milling was carried out in order to establish the relationship between depth and chemical bond intensity. The distances referred to in the paper are measured from $d = 0$.

An Instron testing machine was used to measure the failure load of the LAMP and UAL specimens. Testing was conducted with a crosshead speed of 0.5 mm/min, and the tensile load and displacement curve was recorded. A series of tests was conducted to study the effects of the various laser processing parameters on the load-displacement curve, and the fractured LAMP and UAL joined specimens were examined. For each joining condition, three tensile specimens were tested.

3. Results and discussion

3.1. Joints produced with blind holes in the transducer

Fig. 6 shows the joint appearance of a typical LAMP specimen, i.e. without ultrasonic vibration. Similar to previous findings in other LAMP studies (Chan and Smith, 2016; Tillmann et al., 2010), bubbles are present in the joint zone along the centre part of the weld where the laser power is considerably higher, and decomposition of the polymer occurred. Also the higher the power, the more bubbles formed (Chan and Smith, 2016; Tillmann et al., 2010). Turning to the experiment of using the blind-hole-design tool, it is envisaged that under the action of ultrasonic vibration, a surface cavity in the tool will cause a low

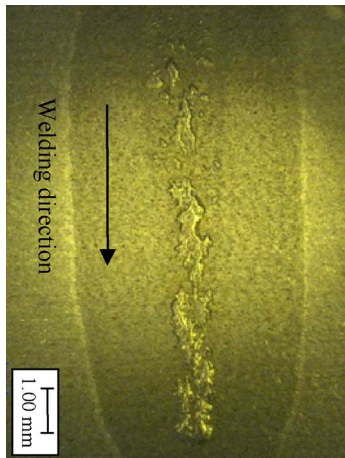


Fig. 6. Joint appearance of the sample produced using LAMP, showing the bubbles developed along the weld centreline region (laser power 55 W, frequency 20 Hz and joining speed 4 mm/s).

pressure region to be developed in the molten plastic zone and this may lead bubbles to be collected in a designated region. The results obtained agree well with our expectation in which most of the bubbles were collected and coalesced into pockets of voids (Fig. 7). The locations of these pockets match the positions of the holes in the tool surface. The processing parameters of this specimen were exactly the same as that used for the LAMP specimen (Fig. 6). The following section examines the effect of pressure induced in the workpiece on the bubble translation path.

3.2. The effect of pressure on bubble translation path

In order to analyze the effect of surface features of an ultrasonic tool on the movement of bubbles, it is necessary to study the pressure generated in the molten plastic by the ultrasonic tool. Applications of ultrasonic waves often require transmission from one medium to another where their effects can be seen. Considering the case of an ultrasonic transducer tool with a surface structure having 3-blind holes (Fig. 2), at Position A of the workpiece which is relatively far from the location of a surface hole in the y-axis, and Position B, which is located directly above the opening of a surface blind hole in the transducer. The transmission and reflection of ultrasonic waves from the transducer to the workpiece at these two locations are different, and cause a pressure difference between these two locations.

3.2.1. Pressure at position A

At position A, the effects of transmission and reflection of ultrasonic waves take place at the interface between the transducer tool and the metal specimen, where the transducer is sending out a plane pressure wave towards the Ti sheet. The acoustic properties of the transducer tool and Ti sheet are different, partial reflection at the interface is expected with a certain degree transmitted through to the Ti sheet (Fig. 8). The incident waveform (P_i), the reflected waveform (P_r), and the transmitted waveform (P_t), can be represented as (David and Cheeke, 2002):

$$P_i = A \exp j(\omega t - k_{tool} z) \quad (1)$$

$$P_r = AR_{tool-Ti} \exp j(\omega t + k_{tool} z) \quad (2)$$

$$P_t = AT_{tool-Ti} \exp j(\omega t - k_{Ti} z) \quad (3)$$

Where A is the vibration amplitude, $j = \sqrt{-1}$, z is displacement, k_{tool} and k_{Ti} are the stiffness coefficients of the transducer tool and the Ti sheet, respectively. ω is angular frequency, $R_{tool-Ti}$ and $T_{tool-Ti}$ are the pressure reflection and transmission coefficients between the transducer tool and the Ti sheet, respectively.

It is assumed that the transducer tool and Ti sheet are in intimate contact and the boundary conditions of pressure and velocity (displacement) at $z = 0$ at the interface between the tool and the Ti sheet remain the same. Based on the definition of acoustic impedance (David and Cheeke, 2002):

$$1 + R_{tool-Ti} = T_{tool-Ti} \quad (4)$$

$$(1 - R_{tool-Ti})/Z_{tool} = T_{tool-Ti}/Z_{Ti} \quad (5)$$

where, Z_{tool} and Z_{Ti} are the characteristic acoustic impedances of the transducer tool and the concerned Ti sheet. Using Eqs. (4) and (5), the pressure transmission coefficient and the reflection coefficient can be determined:

$$T_{tool-Ti} = 2Z_{Ti}/(Z_{Ti} + Z_{tool}) \quad (6)$$

$$R_{tool-Ti} = (Z_{Ti} - Z_{tool})/(Z_{Ti} + Z_{tool}) \quad (7)$$

Where, $Z_{tool} = \rho_1 c_1$, $Z_{Ti} = \rho_2 c_2$; and ρ_1 , ρ_2 and c_1 , c_2 , are the densities and sound propagation velocities of the transducer tool and the Ti sheet, respectively, which are given in Table 3.

At normal incidence, the intensity of the sound wave travelling in a

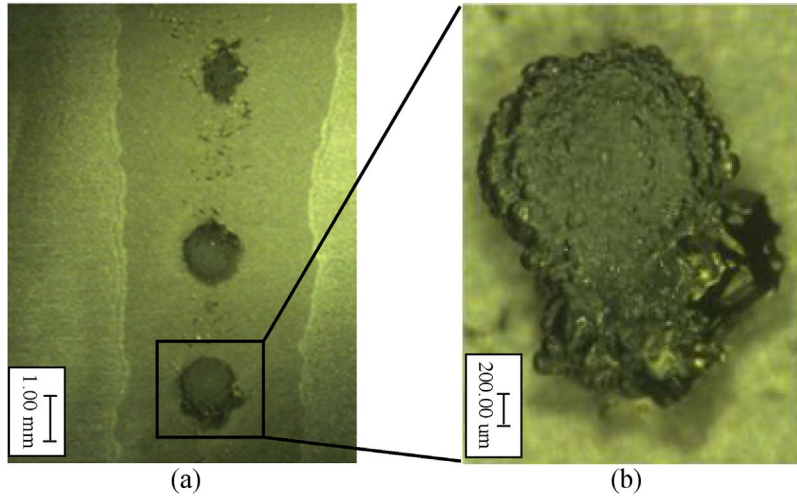


Fig. 7. Joint appearance of the sample produced using a tool surface with 3 circular holes, (a) showing the bubbles collected at regions above the holes; (b) an enlargement of one of these regions (laser power 55 W, frequency 20 Hz and joining speed 4 mm/s).

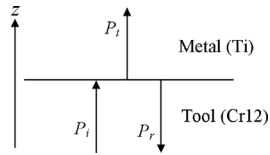


Fig. 8. Reflection and transmission at the interface between transducer tool and Ti sheet.

Table 3
Density and sound velocity of the media.

Media	Density (ρ) (kg/m ³)	Sound velocity (c) (m/sec)
Cr12 steel	7870	5948
Ti	4500	6069
air	1.19	330

material can be expressed as ():

$$I \equiv P_{\max}^2 / 2Z \quad (8)$$

$P_{\max}$ is the maximum pressure and can be expressed as (David and Cheeke, 2002):

$$P_{\max} = 2\pi A f \rho c \quad (9)$$

where, f is the vibration frequency, ρ is the density of the material, c is the velocity of the ultrasound. In the case of UAL, the transducer sends out a plane pressure wave towards the Ti sheet, and the maximum incident pressure at the transducer tool surface ($P_{\max \text{ tool}}$) can be expressed as:

$$P_{\max \text{ tool}} = 2\pi A f \rho_1 c_1 \quad (10)$$

The acoustic intensity of transmission between the transducer tool and Ti sheet can be conveniently determined based on the definition of acoustic intensity (),

$$I_{Ti} = Z_{Ti} / Z_{Ti} \times |T_{tool-Ti}|^2 \times I_{tool} \quad (11)$$

By Eqs. (8), (10) and (11), the maximum transmission pressure that is transmitted to the Ti sheet ($P_{\max Ti}$) is given by:

$$P_{\max Ti} = T_{tool-Ti} \times P_{\max \text{ tool}} \quad (12)$$

Since the thickness of the Ti sheet (1 mm) is much smaller than the length of the transducer (213 mm), the thickness of the Ti sheet can be ignored, and the Ti sheet can be considered to be vibrated in phase with the transducer. Combining Eqs. (10) and (12), the maximum pressure exerted on the molten plastic through the Ti sheet at position A can be expressed as:

$$P_{\max A} = P_{\max Ti} = T_{tool-Ti} \times P_{\max \text{ tool}} = 2\pi T_{tool-Ti} A f \rho_1 c_1 \quad (13)$$

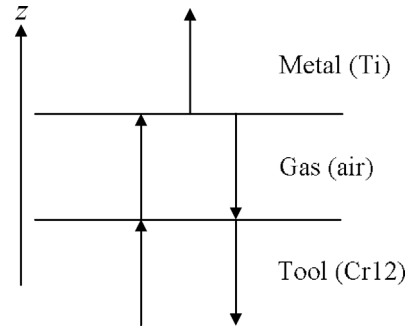


Fig. 9. Reflection and transmission at the interfaces of the compounded layer (the transducer tool, air and Ti sheet).

3.2.2. Pressure at position B

Consider position B, which is located above a blind hole in the transducer tool. In this case, the ultrasonic wave is transmitted from the transducer, through the air, to the Ti sheet (Fig. 9). The multiple reflection and transmission of acoustic waves in the transducer tool, air and the Ti sheet can be treated as occurring in a compounded layer (Fig. 9).

The transmission coefficient of the compounded layer can be expressed as ():

$$T_{tool-air-Ti} = |4Z_{Ti}Z_{air}(\cos \psi - i \sin \psi) / (Z_{Ti} - Z_{air})(Z_{air} - Z_{tool}) + (Z_{Ti} + Z_{air})(Z_{air} + Z_{tool})(\cos \psi - i \sin \psi)^2| \quad (14)$$

Where, Z_{air} is the characteristic acoustic impedance of air, and ψ is the phase advance of the plane wave passing through the air, which can be determined by ():

$$\psi = 2\pi f d / c_3 \quad (15)$$

Where, d is the depth of the holes (air) in the transducer tool as shown in Fig. 2, c_3 is the sound velocity in air, and in this case, $\psi = \pi/2$. (Substitute $f = 28 \text{ kHz}$, $d = 15 \text{ mm}$, $c_3 = 330 \text{ m/sec}$ into Eq. (15) yields $\psi \approx 2.5\pi$. The cycle function of $\cos x$ and $\sin x$ is 2π , then, in this case, $\psi = \pi/2$). Substituting $\psi = \pi/2$ into Eq. (14) yields

$$T_{tool-air-Ti} = 4Z_{Ti}Z_{air} / (Z_{Ti}^2 + Z_{air}^2 + 2Z_{Ti}Z_{tool}) \quad (16)$$

By Eqs. (10) and (16), the maximum pressure exerted on the molten plastic through the Ti sheet at position B can be expressed as:

$$P_{\max B} = T_{tool-air-Ti} \times P_{\max \text{ tool}} = 2\pi T_{tool-air-Ti} A f \rho_1 c_1 \quad (17)$$

Substituting the data given in Table 2 into Eqs. (6) and (16) yields:

$$T_{tool-Ti} / T_{tool-air-Ti} > 1 \quad (18)$$

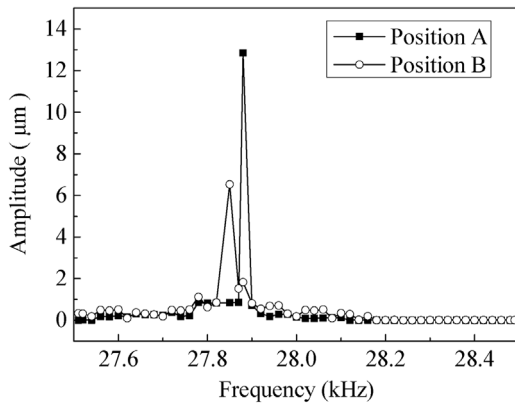


Fig. 10. Vibration amplitude of Position A (far from the hole), and Position B (just above a surface hole).

Hence,

$$P_{\max A} > P_{\max B} \quad (19)$$

The above analysis shows that a surface cavity in the transducer causes a corresponding low pressure region to be developed in the molten plastic. This induced pressure change in the molten plastic will have an effect on the movement of the laser-induced vapour bubbles.

3.2.3. Verification of pressure change due to surface cavities

To verify that the surface holes in the transducer tool can change the pressure acting on the molten plastic, the vibration amplitude at the two locations, Position A and Position B, were measured using a laser Doppler vibrometer (Polytec, PSV-400) (Fig. 4). The displacement measurements for these two positions are given in Fig. 10. The measured vibration amplitude of Position A was about two times that of Position B. The vibration amplitude at position A is greater than that at Position B, according to equation (9), the pressure acting at Position A should be greater than that acting at Position B, i.e. $P_{\max A} > P_{\max B}$. This would cause a pressure difference in the molten plastic. Bubbles in the relatively high pressure regions (without a cavity underneath) will migrate to the relatively low pressure regions (with a cavity underneath). As a result, they are collected at Position B to form a relatively large bubble. The pressure analysis results presented in Sections 3.2.1 and 3.2.2 are in agreement with the experimental results obtained (Fig. 7), where bubbles accumulated at positions above the surface hole regions due to a relatively low pressure existing there.

The results obtained so far have shown that under the influence of ultrasonic vibration, a cavity in the transducer tool creates a low pressure region in the specimen above the location of a cavity. A similar analysis for the feature of a channel in the ultrasonic tool also shows that a low pressure region is present above the open channel.

3.3. Joints produced with a transducer with an open channel

For both LAMP and UAL processes, laser-induced bubbles were developed along the centreline region of the joint where laser power was relatively high (Fig. 6). For UAL, the advantage of a pressure differential across the surface of a transducer with an open channel is to drive bubbles away and to escape from the molten pool of the joint. According to the analysis given in the previous section, if the open channel is placed offset from the centreline of the joint, the bubbles could migrate from the central region of high pressure to a region of low pressure where the channel is lying underneath. Fig. 11 shows that if the edge of the open channel is in line with the centreline of the joint, bubbles can be forced to move to one side of the molten zone and made to escape through the PET/Ti interface. In the case where there is no offset (Fig. 12(a)) or the offset is too large (Fig. 12(b)), there will not be a large pressure difference around the bubbles. Therefore, the bubbles

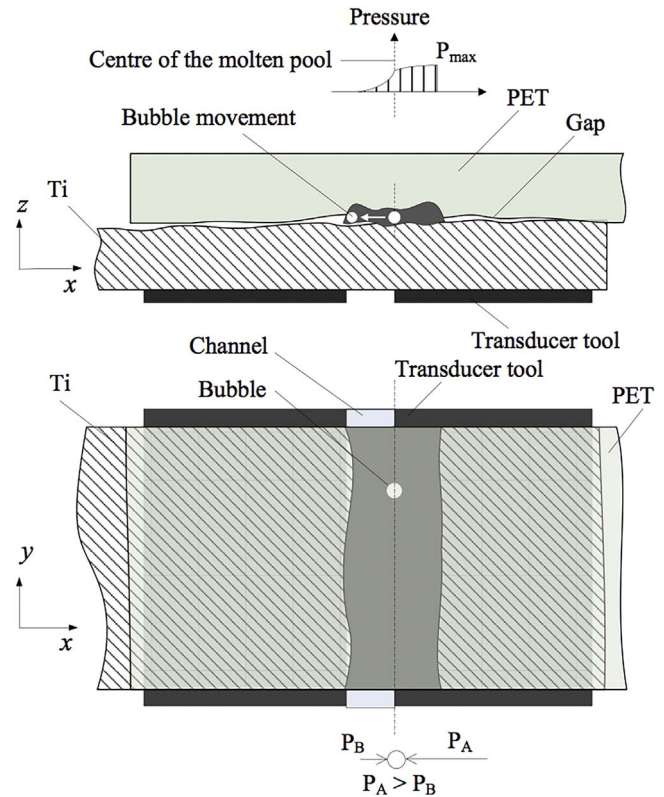


Fig. 11. A schematic diagram showing the effect of pressure difference on the movement of bubbles.

cannot easily escape.

A series of experiments were conducted to verify the postulation in Fig. 11 at different laser powers. Fig. 13 shows the appearance of the joints produced using an open channel transducer offset 0.5 mm to the left of the centreline of the weld (i.e. Fig. 11). The results show that the relatively large bubbles formed in the LAMP specimens disappeared when the transducer was offset to one side with the open channel lined up with the edge of the joint (Fig. 13), and only few small pores were found in the UAL specimens (Fig. 14). Fig. 13 also shows a comparison of the specimens produced using LAMP joining, i.e. without ultrasound applied but using the same processing parameters.

In comparing the UAL and the LAMP joined specimens, very encouraging results were obtained for the UAL process. With the new ultrasonic-aided laser joining method – using the conjoint action of a laser and ultrasonic – and the offset technique, the amount of bubbles was significantly reduced and in most cases no sizable bubbles could be observed under an optical microscope. An examination of the cross-sections of the joined specimens showed that many bubbles were trapped at the interfaces of the joints of the LAMP specimens, while hardly any were found in the UAL joined specimens (Fig. 15). The elimination of bubbles at the joint zone increases the joint strength.

3.4. Failure study of joints

Fig. 16a shows the effect of laser power on failure load for both LAMP and UAL specimens. The area of bonded surface of the joint was measured before the shear test, the area increases as laser power increases (Fig. 17). To better reflect joint performance, Fig. 16a was re-plotted in terms of failure stress (Fig. 16b). Typical fracture appearance of the LAMP and UAL specimens are shown in Fig. 18. The former fractured at the metal-plastic interface and resulted in separation of the metal and plastic parts, while the latter fractured in the parent plastic part. It is clear that fracture stress of the UAL specimen is significantly higher than that of the LAMP specimens (at least four times higher) and

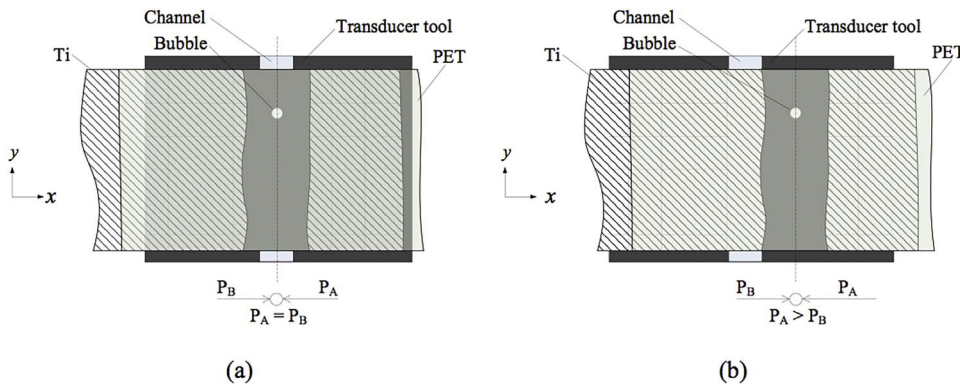


Fig. 12. A schematic diagram showing the pressure acting on a bubble (a) without offset of the transducer; (b) with a large offset of the transducer.

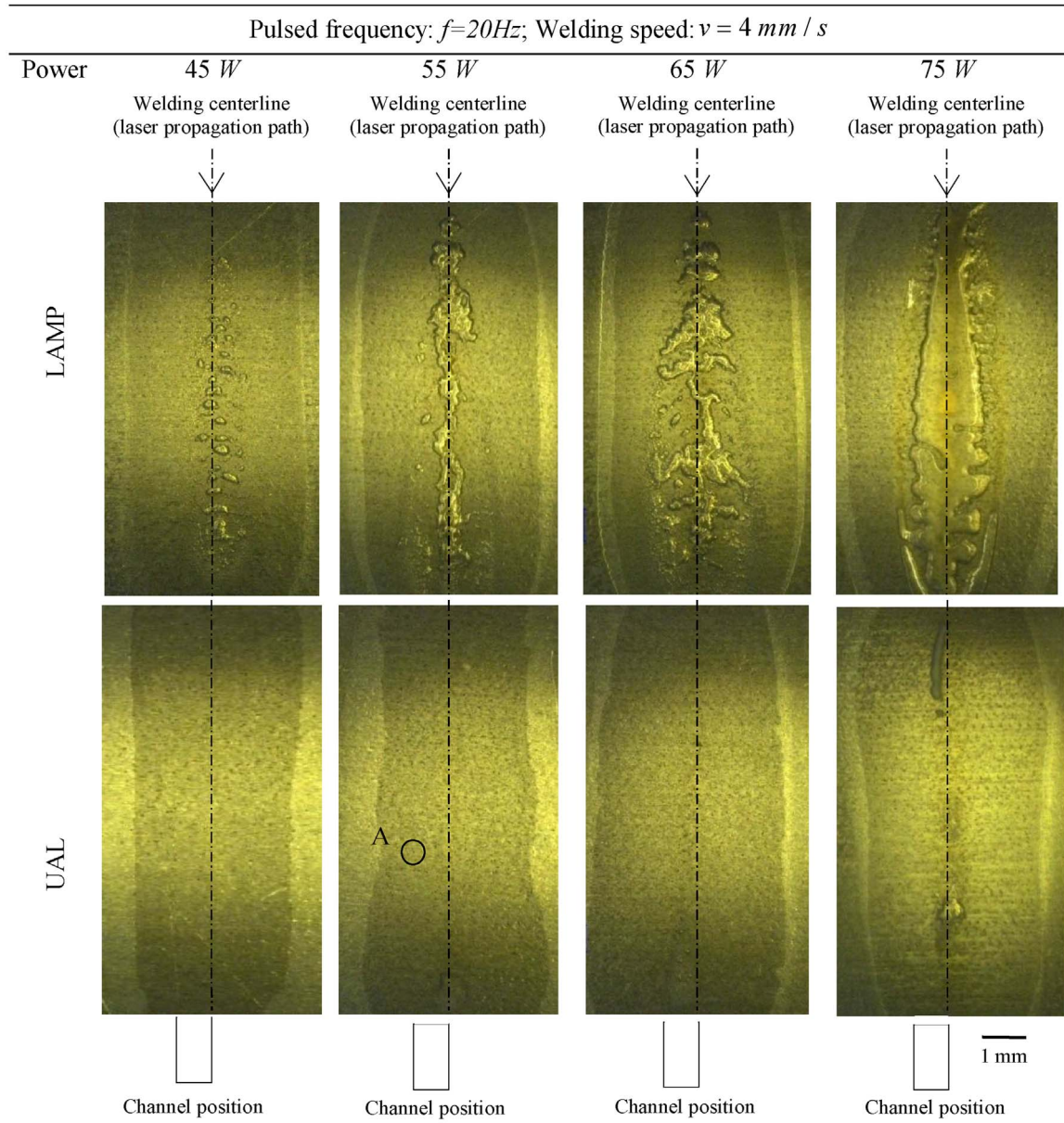


Fig. 13. Typical joint appearances as a function of laser power for both LAMP and UAL (with offset of 0.5 mm) joined specimens.

their fracture mechanisms were different. This is considered to be caused by a reduction of bubbles and a stronger metal-plastic interface obtained for the UAL specimens. This strong interface of the UAL specimens is supported by a higher level of Ti–C bond intensity and a

greater depth of reaction zone of Ti–C bond measured across the interface. The results obtained for the LAMP and UAL specimens are given in Fig. 19. A more detailed analysis of the chemical bonds formed at the Ti/PET interface can be found in the previous study (Chen et al., 2016).

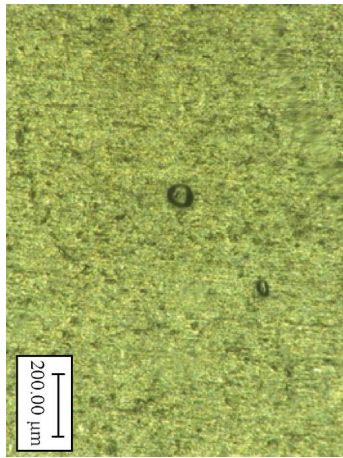
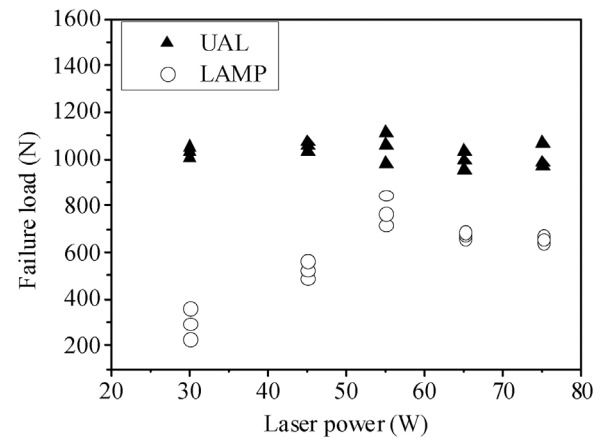


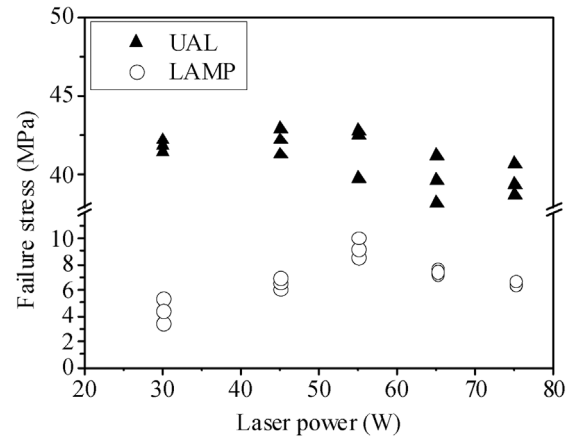
Fig. 14. Only few small bubbles can be found in a UAL specimen (Fig. 13) at location A.

For the LAMP specimens, the failure stress initially increases with increase in laser power and reached a maximum at 55 W. It then decreased when the power was increased further (Fig. 16b). The level of laser power is considered to have a direct effect on the formation of chemical bonds at the joint interface, as well as on polymer degradation. Too low a laser power will reduce chemical bonding between the metal and plastic parts, whereas, too high will result in the formation of many bubbles, as well as causing degradation of the polymer, thus resulting in a relatively weak joint. Fig. 20 reveals that the final fracture area of LAMP_55 W was much larger than that of a LAMP specimen produced using lower laser power (LAMP_30 W) (cf. Figs. 20 a & 21 a), with a high degree of plastic drawing (cf. Figs. 20 b & Fig. 21b). This leads to the conclusion that bubble-induced pressure acting on the molten plastic does help to form a stronger bond at the interface. However, if too many bubbles are formed when high laser powers are used, the counter-effect of bubbles (themselves as defects) becomes significant.

Examining the results of UAL specimens, the failure stress did not change significantly within the range of laser power used (i.e. 30 W–75 W), except that when laser power was above 55 W, fracture strength slightly decreased. This indicates that when the laser-bonded joint is largely free from porosity, increasing laser power will not result in higher strength, and if the power is too high, it may even cause a reduction in strength due to damage made to the plastic. Fig. 22 shows that fracture of the UAL specimens occurred in the parent plastic material (specimens UAL_30 W and UAL_55 W), and the final bonded area between the PET and the Ti sheets was larger for the latter. For UAL_30 W, a relatively large part of the originally bonded surface has already debonded. This indicates that, although the overall interfacial bond strength of UAL_55 W was higher than that of UAL_30 W,



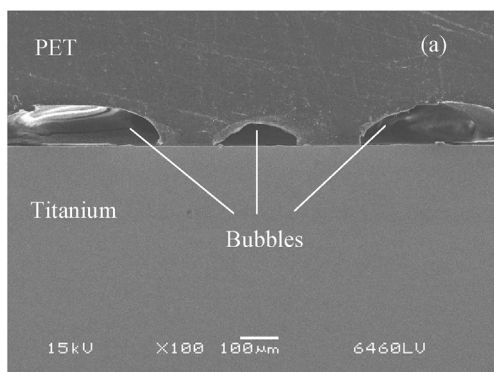
(a)



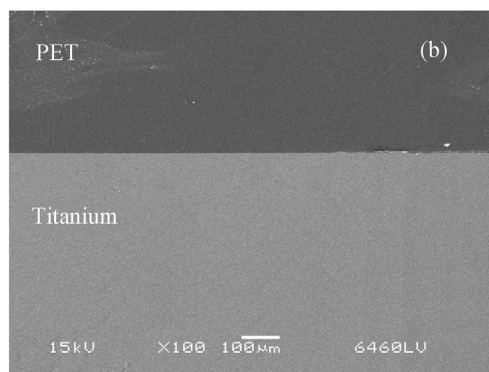
(b)

Fig. 16. The effects of laser power on (a) failure load, (b) failure stress.

especially in regions further away from the centre of the weld line, a high interfacial bond still occurs near the joint centreline of UAL_30 W. This is reasonable because laser output has a Gaussian distribution and the highest power is experienced at the centre of the weld. This means that towards the end of the tensile shear test, the remaining bonded surface along the centreline has sufficient interfacial strength to hold the PET and Ti parts together and eventually the fracture occurred in the plastic part. This is considered to be the reason why the failure stress did not change considerably with laser power.



(a)



(b)

Fig. 15. Cross-sections of the joints, showing the joint interfaces of the (a) LAMP specimen and (b) the UAL joined specimen (laser power 55 W, frequency 20 Hz and joining speed 4 mm/s).

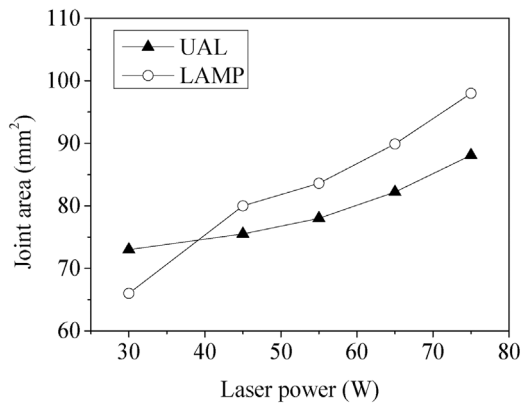


Fig. 17. The relationship between laser power and joint area.

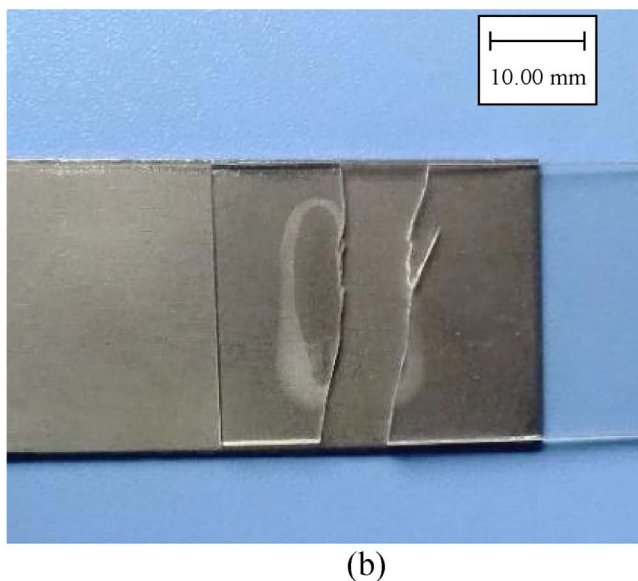
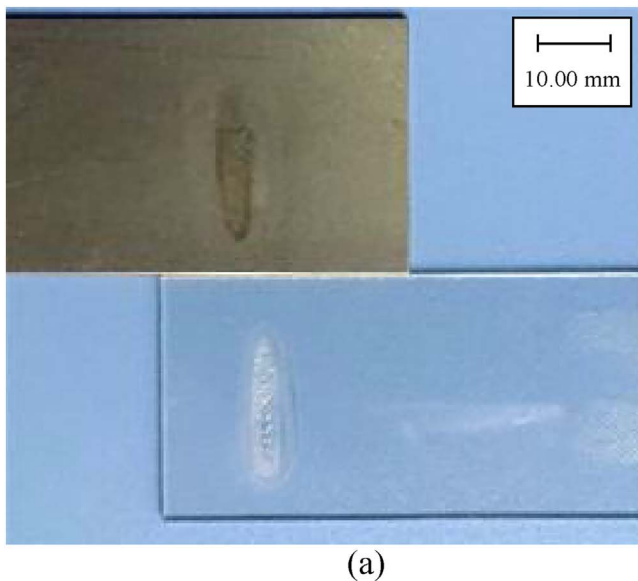
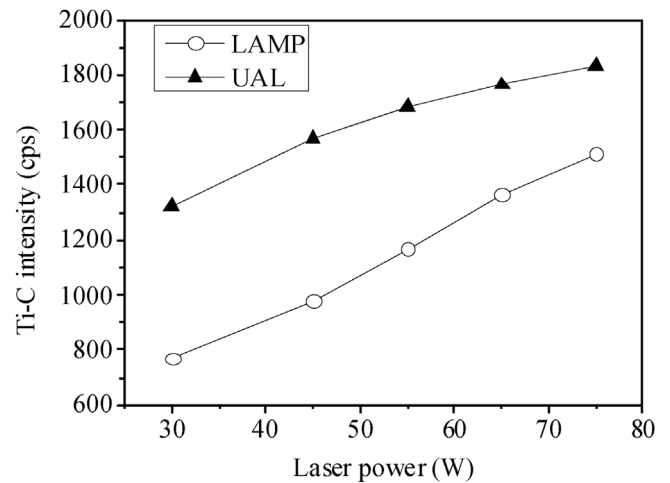


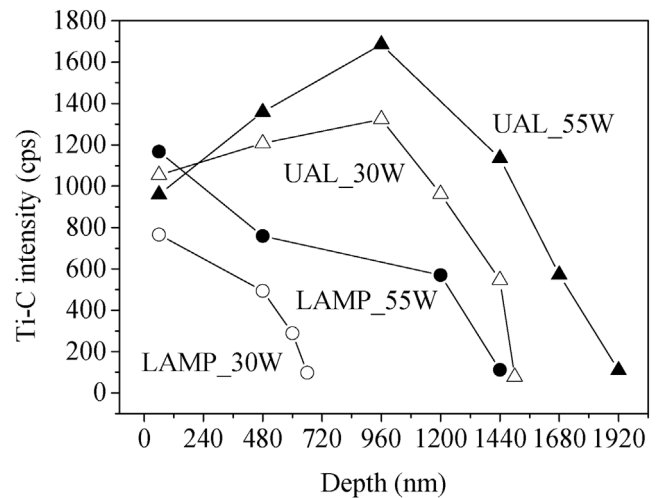
Fig. 18. Failure appearance of (a) a LAMP specimen, (b) a UAL specimen.

4. Conclusions

- (i) For the conventional LAMP joining process, the formation of bubbles in the joint zone of Ti/PET sheets is inevitable and the



(a)



(b)

Fig. 19. (a) The maximum Ti-C intensity measured for the various LAMP and UAL specimens, (b) Ti-C intensity measured at different sputtering depths for some LAMP and UAL specimens.

amount increases when laser power is increased. Bubbles can be significantly reduced when ultrasonic vibration is incorporated into LAMP with a properly designed transducer tool. It is considered that UAL can be potentially applied to other metal-plastic pairs to achieve similar outcomes.

- (ii) A transducer tool with a surface cavity is needed to create a low pressure region in the molten plastic for driving bubbles out of the joint, and in this case, an open channel was used.
- (iii) The failure stress of the LAMP joints increases with increase in laser power and reaches a maximum. A further increase in power results in a decrease in strength.
- (iv) For the UAL joints, laser power has a less effect on failure stress, except that when a relatively high power was used (higher than 55 W), a slight reduction in strength began.
- (v) The failure stress of UAL joints was at least four times higher than that of LAMP joints.

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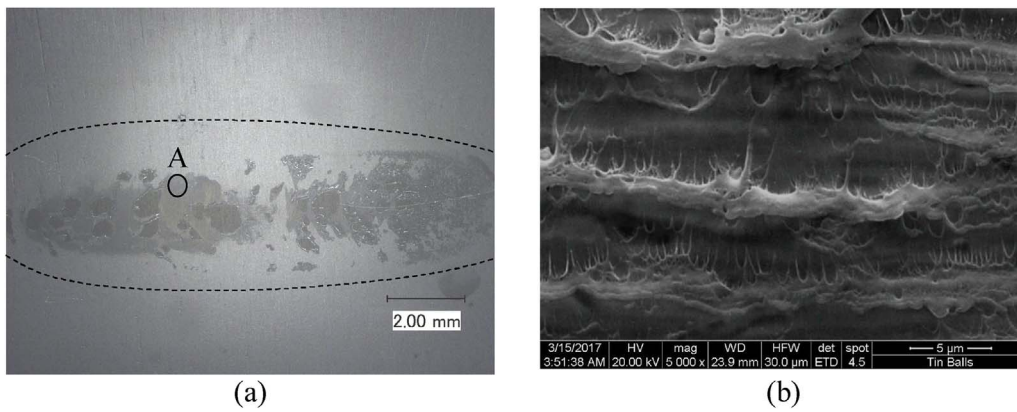


Fig. 20. Fracture surface of LAMP_55 W, photos taken from the Ti side (a) optical micrograph (the dotted line indicates the originally bonded boundary), (b) SEM micrograph of point A.

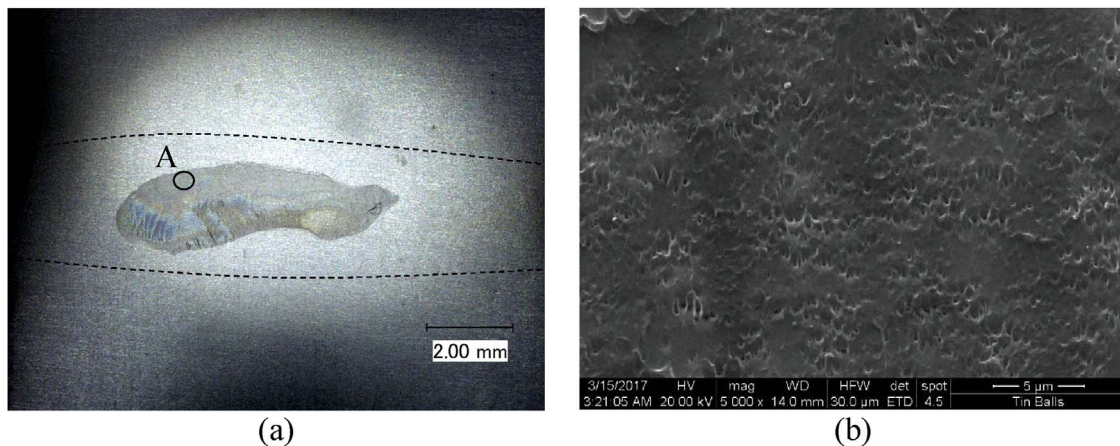


Fig. 21. Fracture surface of LAMP_30 W, photos taken from the Ti side (a) optical micrograph (the dotted line indicates the originally bonded boundary), (b) SEM micrograph of point A.

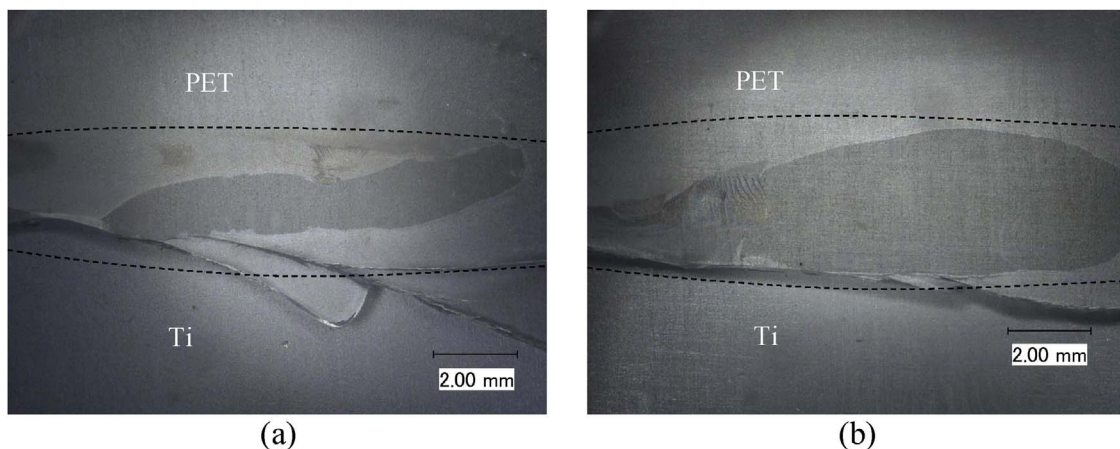


Fig. 22. Fracture occurred in the parent plastic material (a) UAL_30 W, (b) UAL_55 W specimens; (the dotted line indicates the originally bonded boundary).

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